

Option Pricing Analysis: Black-Scholes-Merton vs. Market Prices Using Yahoo Finance Implied Volatility for Selected U.S. Stocks

Abdoulaye Maiga^{*1}

¹School of Mathematics and Statistics, Rochester Institute of Technology, Rochester, NY,
United States

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^{*}Correspondence: am9770@rit.edu

Abstract

This project examines the pricing performance of the Black–Scholes–Merton framework for short-maturity U.S. equity call options using daily market data obtained from Yahoo Finance. The analysis compares theoretical option prices computed with and without dividend adjustments and evaluates their alignment with observed market prices. By focusing on a transparent implementation and a narrow time window, the study highlights the practical role of dividend assumptions in applied option pricing.

1 Introduction

The Black–Scholes option pricing model remains a foundational tool for valuing equity options due to its analytical tractability and widespread use in both academic and market settings [1]. Although the model relies on simplifying assumptions such as constant volatility and frictionless markets, it continues to serve as a benchmark against which observed option prices are commonly evaluated.

Merton [2] extended the original framework to account for dividend-paying assets, highlighting the importance of dividend treatment in equity option valuation. In practice, dividends reduce the expected future value of the underlying stock and can materially affect call option prices, particularly for short-maturity contracts. Empirical evidence further shows that deviations between market prices and Black–Scholes predictions are systematic rather than random, with implied volatility varying across strikes and maturities [3].

Motivated by these findings, this project applies the Black–Scholes–Merton framework to short-maturity U.S. equity call options using daily market data obtained from Yahoo Finance. The analysis compares theoretical prices computed with and without dividend adjustments and examines their alignment with observed market prices. By focusing on a transparent implementation and a narrow time window, the project highlights the practical impact of dividend assumptions on option pricing accuracy.

2 Literature review

The Black–Scholes model introduced by Black and Scholes [1] provides a closed-form solution for pricing European-style options under a set of simplifying assumptions, including constant volatility, frictionless markets, and lognormally distributed asset returns. Despite these assumptions, the model remains a central benchmark in option pricing due to its analytical tractability and its continued use in both academic research and market practice.

Merton [2] extended the original Black–Scholes framework to incorporate dividend-paying assets and relaxed certain assumptions, allowing the model to be applied more broadly in equity option markets. In particular, dividend adjustments play a critical role in pricing equity call options, as expected dividend payments reduce the forward value of the underlying stock. In applied settings, dividends are commonly incorporated through a continuous dividend yield, preserving the simplicity of the Black–Scholes framework while accounting for expected cash flows.

A substantial body of empirical evidence documents systematic deviations between observed option prices and those predicted by the Black–Scholes model. One of the most prominent features is

the implied volatility smile, where implied volatility varies across strike prices and maturities, contradicting the model’s constant-volatility assumption. Rubinstein [3] shows that such patterns are persistent and economically meaningful, suggesting that pricing discrepancies often reflect model simplifications rather than arbitrage opportunities.

3 Methodology

3.1 Research methodology

This study applies the Black–Scholes–Merton (BSM) framework to short-maturity U.S. equity call options and compares theoretical prices with observed market prices. The analysis focuses on the impact of dividend assumptions by computing option prices under two specifications: (i) the standard Black–Scholes model without dividends and (ii) the Black–Scholes–Merton model with a continuous dividend yield.

For each trading day in the sample, call option data are collected for a selected set of U.S. equity tickers. For each option contract, the observed market premium, strike price, time to maturity, and implied volatility are obtained from Yahoo Finance. The underlying stock price corresponds to the closing price on the observation date. Time to maturity is expressed in years using a day-count convention based on calendar days remaining until expiration.

The risk-free interest rate is assumed to be constant over the short sample window and is treated as exogenous. Implied volatility provided by Yahoo Finance is used as the volatility input to the pricing formulas, allowing the analysis to isolate pricing differences arising from dividend treatment rather than volatility estimation.

Using these inputs, theoretical call option prices are computed under both pricing specifications for each contract. The resulting model prices are then compared to observed market prices to assess pricing differences across tickers, strikes, and trading days. The analysis emphasizes transparency and consistency of implementation rather than statistical hypothesis testing, given the short time horizon of the dataset.

3.2 Data

The option data used in this study are obtained from Yahoo Finance and consist of daily snapshots of equity call option chains for a selected group of U.S. stocks. The sample spans multiple trading days over a two-week period in September, with separate data files collected for each observation date. Additional files corresponding to subsequent trading days follow the same structure and variable definitions.

Each dataset contains call option contracts for a single expiration date and includes, among other variables, the strike price, last traded option price, bid and ask quotes, trading volume, open interest, and implied volatility. For each observation date, the corresponding underlying stock price and dividend information are also recorded. Dividend yields are obtained directly from Yahoo Finance and are used to construct the continuous dividend adjustment in the pricing model.

The individual daily datasets are consolidated into a unified format to facilitate consistent pricing and comparison across dates. All option contracts with missing or invalid pricing inputs are excluded from the analysis. Given the short maturity and limited time span of the data, the study

focuses on pricing behavior across contracts rather than long-term dynamics.

All datasets are stored in comma-separated value (CSV) format and processed using a reproducible workflow, allowing the pricing results to be replicated using publicly available data sources.

4 Empirical results

4.1 Descriptive overview

The empirical analysis is based on daily snapshots of U.S. equity call option chains collected over a three-week period. For each trading day, option contracts are observed across multiple strike prices for a fixed expiration date, resulting in a cross-section of short-maturity call options for each underlying stock.

Across the sample, option prices vary systematically with strike and underlying price, with the highest premiums observed for in-the-money contracts and lower prices for out-of-the-money options. Implied volatility levels obtained from Yahoo Finance exhibit moderate variation across strikes, consistent with the presence of an implied volatility smile. Dividend yields differ across the selected stocks, providing a natural setting to evaluate the impact of dividend adjustments in the pricing model.

Table 1: Summary statistics by ticker

Ticker	# Contracts	Avg Call Price	Avg Impl. Vol.	Avg Div. Yield	MAE (No Div.)	MAE (With Div.)
NVDA	3037	141.1578	3.8630	0.0002	4.9174	4.9167
MSFT	1304	72.9070	0.8967	0.0066	2.2151	2.1573
UNH	1276	36.5766	1.3029	0.0285	5.8272	5.6989
AAPL	1061	66.6472	2.1249	0.0045	1.1801	1.1581
JPM	839	57.0868	0.8266	0.0186	7.3160	7.1882
BAC	708	8.3648	1.0454	0.0221	0.8328	0.8072
WFC	637	12.1893	0.8077	0.0219	0.8252	0.7888
MRK	620	5.5061	1.0283	0.0385	0.6766	0.6366
PFE	477	2.6348	1.1319	0.0695	0.1737	0.1509

Notes: Summary statistics are computed by pooling all observed call option contracts for each ticker across trading days. Contracts refer to daily option observations; the set of available strikes varies across dates as contracts enter and expire.

4.2 Market prices versus model prices

This section compares observed market call option prices with theoretical prices generated by the Black–Scholes–Merton framework with dividend adjustments. Model prices are computed for option contracts observed across thirteen trading days. For visual clarity, figures are presented for a representative trading day, while results are qualitatively similar across the remaining days in the sample.

Figure 1 presents a scatter plot of observed market prices against dividend-adjusted Black–Scholes model prices for the trading day of August 29, 2025.

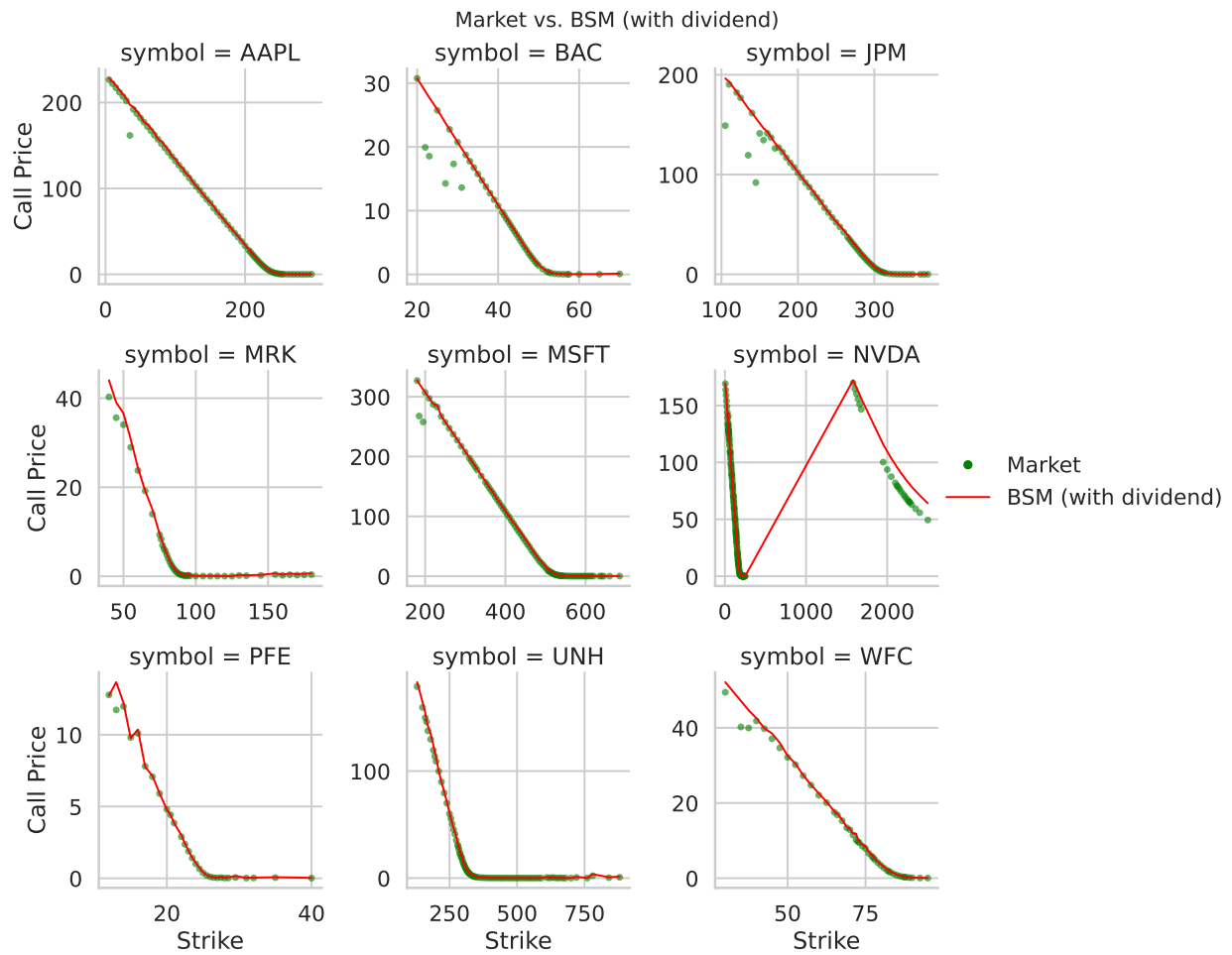


Figure 1: Observed market call option prices versus dividend-adjusted Black–Scholes model prices on August 29, 2025. Each subplot corresponds to a different underlying stock.

Figure 2 reports the same comparison for a different trading day (September 9, 2025) within the sample, illustrating that the relationship between model prices and observed market prices remains stable across trading days.

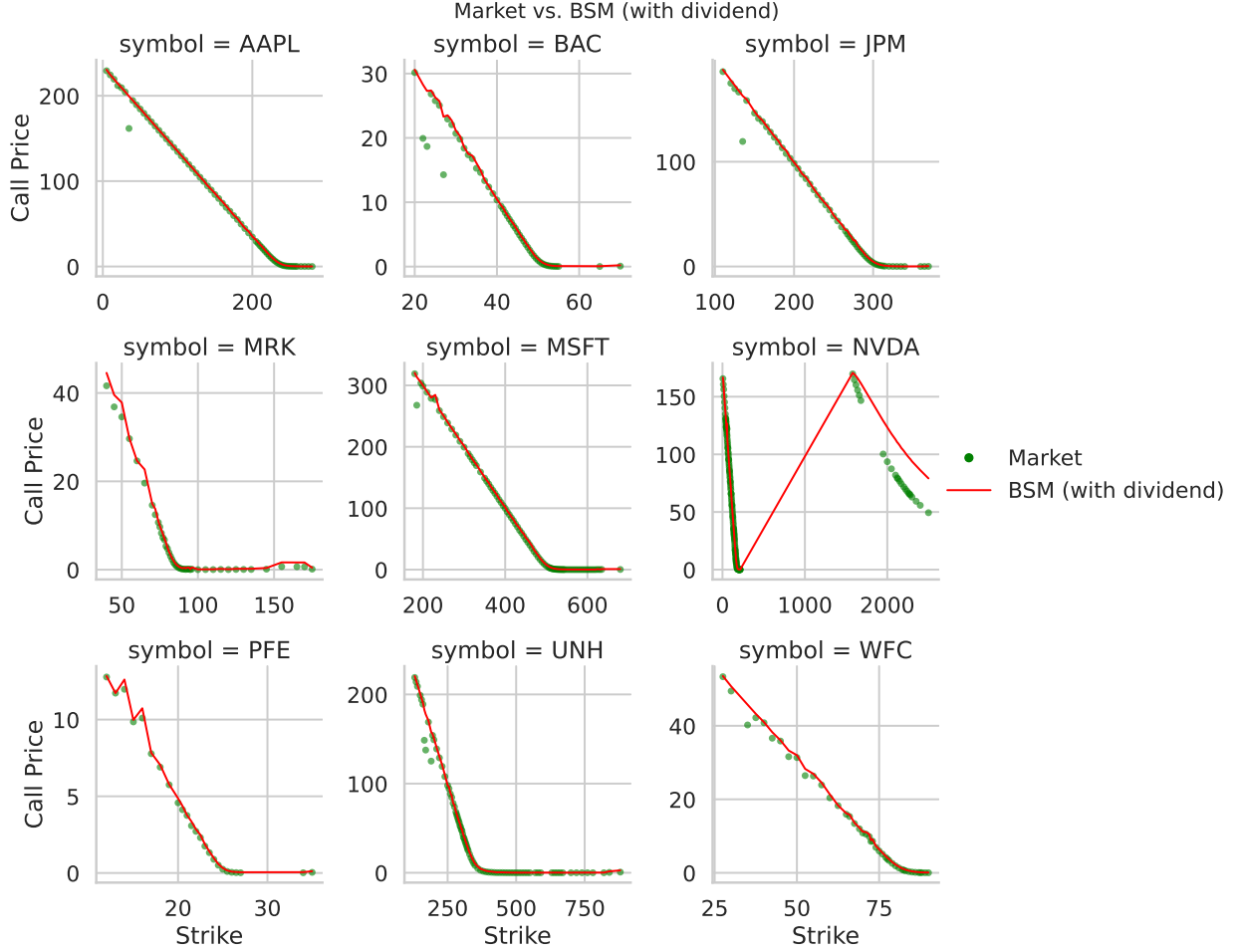


Figure 2: Observed market call option prices versus dividend-adjusted Black–Scholes model prices on September 9, 2025. Each subplot corresponds to a different underlying stock.

Both figures show strong positive relationships between model prices and market prices, indicating that the model captures the overall structure of call option pricing. Deviations from market prices are more pronounced for deep in-the-money and deep out-of-the-money contracts, consistent with known limitations of the constant-volatility assumption.

4.3 Pricing errors and the role of dividends

This section evaluates pricing accuracy by comparing the relative pricing errors of the Black–Scholes model without dividends and the dividend-adjusted Black–Scholes–Merton model. Without loss of generality, a representative trading day is chosen as August 29, 2025.

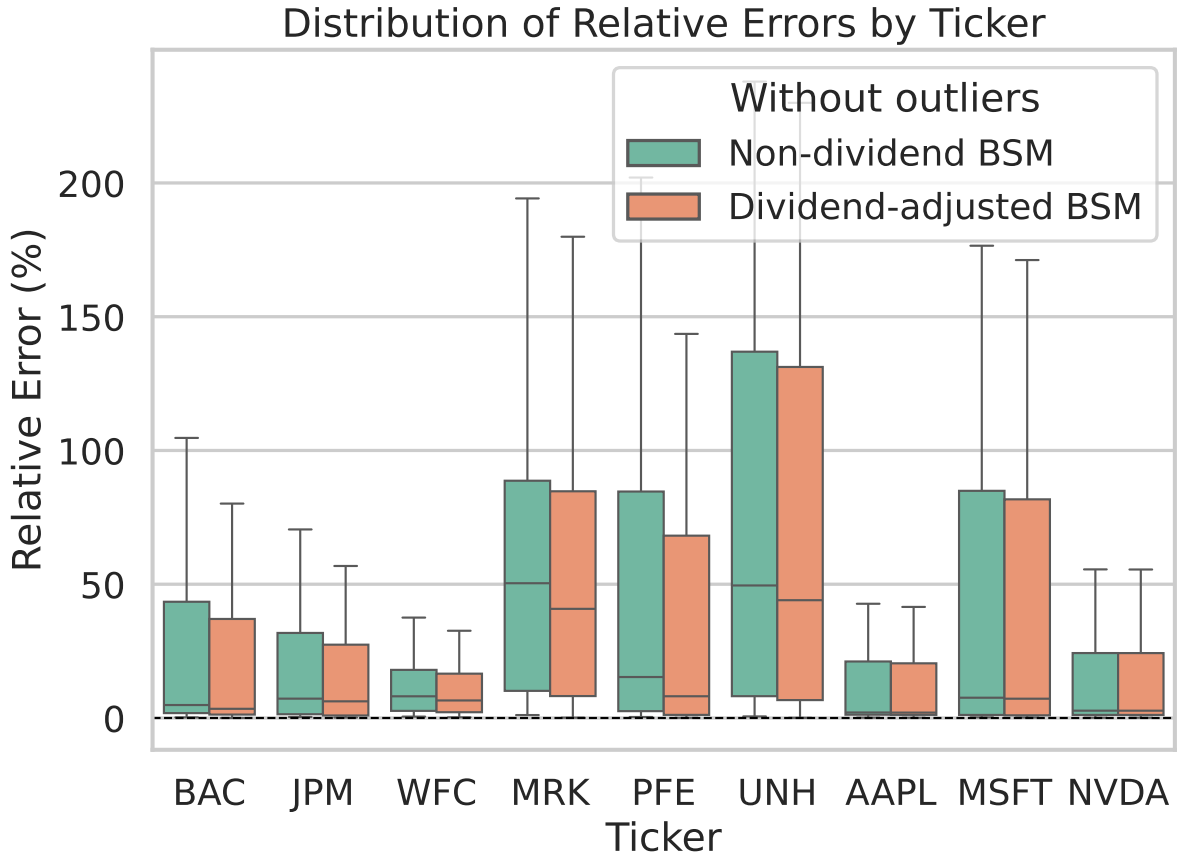


Figure 3: Distribution of relative pricing errors by ticker for the Black–Scholes model without dividends and the dividend-adjusted Black–Scholes–Merton model. Relative errors are computed at the contract level and for the trading day of August 29, 2025. Outliers are removed for clarity.

Figure 3 displays the distribution of relative pricing errors by ticker for both model specifications, with outliers removed for clarity. Across most stocks, the dividend-adjusted model exhibits lower median relative errors and reduced dispersion compared to the non-dividend specification. The improvement is particularly pronounced for dividend-paying stocks such as BAC, JPM, MRK, PFE, and UNH, where ignoring dividends leads to systematically larger pricing errors.

For low-dividend stocks such as NVDA and AAPL, differences between the two models are smaller, indicating that dividend adjustments play a limited role when dividend yields are negligible. Nevertheless, even for these stocks, the dividend-adjusted model does not perform worse than the non-dividend specification.

Despite the overall improvement from dividend adjustments, substantial pricing errors remain for certain stocks, notably UNH. This suggests that while incorporating dividends improves pricing accuracy, remaining discrepancies are driven by other simplifying assumptions of the Black–Scholes framework, such as constant volatility and continuous trading.

5 Discussion and Conclusion

This project examined the empirical performance of the Black–Scholes–Merton option pricing framework for short-maturity U.S. equity call options using market data collected across thirteen trading days. The analysis focused on evaluating model fit and assessing the role of dividend adjustments in explaining observed option prices.

The results indicate that the dividend-adjusted Black–Scholes–Merton model provides a reasonable approximation to observed market prices across a range of underlying stocks. Scatter plot comparisons show a strong positive relationship between model prices and market prices, with consistent patterns observed across trading days. Pricing errors are generally larger for deep in-the-money and deep out-of-the-money contracts, reflecting known limitations of the constant-volatility assumption.

Comparisons between dividend-adjusted and non-dividend model specifications reveal that accounting for dividends improves pricing accuracy for dividend-paying stocks. Relative pricing errors are lower and less dispersed when dividends are incorporated, particularly for financial and healthcare stocks with non-negligible dividend yields. For low-dividend stocks, differences between the two specifications are smaller, indicating that dividend adjustments play a limited role when dividend yields are negligible.

Despite these improvements, substantial pricing errors remain for certain stocks, highlighting the simplifying assumptions inherent in the Black–Scholes framework. In particular, the assumption of constant volatility and the use of a European option pricing model for American-style equity options likely contribute to remaining discrepancies between model and market prices.

Several extensions could build on this analysis. Future work could incorporate longer sample periods, allow for time-varying or stochastic volatility, or compare alternative pricing models such as local volatility or stochastic volatility frameworks. Additionally, extending the analysis to include put options or alternative measures of pricing error could provide further insight into model performance.

Overall, the findings suggest that while the Black–Scholes–Merton model remains a useful benchmark for pricing short-maturity equity options, incorporating dividends is an important practical consideration for improving pricing accuracy.

Data availability statement

Publicly available datasets were analyzed in this study. The data was pulled from Yahoo Finance.

Author contributions

AM conducted all aspects of the research presented in this manuscript, including the study design, data collection, and manuscript writing.

Conflict of interest

This research was conducted in the absence of any commercial or financial relationship that could be construed as a potential conflict of interest.

Supplementary material

The Supplementary material for this article can be found on <https://github.com/abdallahmaiga10/BSM-option-pricing>

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